

Jeff Wu

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Education

2021 – 2025: Imperial College London, UK

PhD, Master of Science (*with Distinction*) in Transport

- Awarded a four-year scholarship by the Economic and Social Research Council (ESRC) through the London Interdisciplinary Social Science Doctoral Training Partnership (LISS DTP).

2011 – 2016: University of Melbourne, Australia

Master of Engineering (*with Distinction*), **Bachelor of Environments** in Civil Engineering

- Awarded: Dean's Honours List (x2), Department of Infrastructure Engineering Best Thesis Runner-up, Engineering Exchange Scholarship, Golder Associates Award, Melbourne Mobility Award, Melbourne Global Scholars Award, Norman Westmore Memorial Prize, Surefoot Innovative Ground Solutions Engineering Award, Tsinghua International Scholarship.

Relevant experience

Policy consultant (2024): Imperial College London, London, UK

LTN24: Commissioned by the London Borough of Haringey to evaluate its Low Traffic Neighbourhoods (LTNs) policy. We used a difference-in-differences research design to estimate the policy's causal impact on air pollution and traffic counts. Our findings directly informed the council's cabinet reports and formed the basis of discussions during public hearings, which ultimately influenced the decision to retain the LTNs in Haringey.

- Estimated causal impacts using two-way fixed effects (TWFE) regression.
- Developed an R programme to extract traffic data (ATC, MCC) from over 100 Excel files.
- Managed and visualised spatial data in QGIS.
- Lead author of the consultancy report (available via: [Haringey committees' meetings and minutes](#)).

Transport modeller (2022 – 2023): VLC, London, UK

TSTM23: Developed a Strategic Transport Model for Thurrock Council, forecasting transport network performance and identifying areas for improvement. The modelling results provided critical evidence to support Thurrock's new Local Plan, which outlines the scale and locations of new developments in the borough through to 2040.

- Created base-year trip matrices from mobile phone network data consisting of 50 billion trips.
- Validated base-year trip matrices against independent data sources (e.g., NTS, NTEM, SERTM).
- Prepared public transport data in GTFS format to identify travel modes in observed trips.
- Authored a consultancy report covering model design and data preparation.

PhD researcher (2021 – 2025): Imperial College London, London, UK

LTN25: Furthered the investigation into the causal impacts of LTNs on air pollution and traffic counts. I enhanced the statistical model and found heterogeneous treatment effects on air pollution. The positive impacts on air quality are more pronounced in the inner areas of the neighbourhoods and diminish towards the boundaries.

- Adopted cluster-robust standard errors to account for temporal correlation in the TWFE regression.

- Censored regression for heavy vehicles to account for frequent zero counts on minor roads.
- Derived spatial attributes of measurement sites and LTNs in QGIS (e.g., shortest distances, areas).
- Presented findings at the 2025 Transportation Research Symposium in Rotterdam.

LS25: Improved the neoclassical labour–leisure model by incorporating labour hours into the utility function, and committed expenses (e.g., commuting costs) & time (e.g., commuting time) into the resource constraints. These additions provide theoretical explanations for empirical observations previously unexplained by the standard model, such as the S-shaped labour supply curve. The new model has rich behavioural implications, offering an advanced framework for predicting changes in labour supply and welfare under various policy scenarios.

HW25: Proposed a microeconomic model of commuting frequency, contracted working hours, and actual working hours. This model provides an advanced framework for predicting (1) how workers with flexible work schemes (e.g., hybrid workers) adapt to changes in commuting costs, commuting time, and wage rates; (2) how workers' behaviour, welfare, and value of time change with different flexible work schemes. I developed a discrete–continuous econometric structure for this model and fitted it to a Swiss time-use dataset. The model estimates (1) the elasticities of commuting frequency and working hours; (2) the values of commuting time and flexible work schemes.

- Estimated a mixed multinomial logit model for commuting frequency and contracted working hours.
- Linear regression model with multinomial selection-bias correction for actual working hours.

TU22: Fitted a mixed MDCEV model to the 2015 UK Time-use Survey. The model revealed the effects of flexible work location and hours on workers' time-use patterns. I also validated the model using a hold-out sample test and demonstrated that it can generate activity episodes with accurate duration and timing for use in activity-based and agent-based microsimulation systems.

- Presented findings at the 2022 UTSG Conference in Edinburgh and 2022 IATBR in Santiago.

Other experience

Market risk analyst (2021): AGL Energy, Melbourne, Australia

Developed and maintained databases, models, simulations, dashboards, and software systems to manage market risks for AGL's gas business. By tracking market prices and AGL's positions, I advised traders on risk exposures and hedging strategies.

Software engineer (2017 – 2021): Axiom Software Laboratories, Sydney & Melbourne, Australia

Implemented Axiom software systems in two of Australia's largest banks to process transaction data into high-level risk and regulatory reports, which were sent to management, regulators, and the central bank. While collaborating with clients on-site, I also designed functional features (e.g., data freezing, sign-off, archiving) and streamlined the software's integration into their business processes.

Relevant skills

Programming (R, MATLAB, Python, VBA, shell, SQL) , QGIS, ArcGIS, PowerBI, Tableau, Git/GitHub, statistical modelling, causal inference, optimisation, text mining and analysis, data engineering.

Qualifications: GARP Financial Risk Manager FRM Certification (2021), Toastmasters Level 4 Certificate in Public Speaking and Coaching (2019), Westpac Agile Software Development (2018), PRINCE2 Foundation Certificate in Project Management (2017), BCS Foundation Certificate in Business Analysis (2017).